

Lecture 9. Correlations

We already know that a variable is, by definition, a factor that varies. We have already encountered measures of the *variance* within a single variable. With correlations, we're interested in whether two variables vary together in a systematic way (i.e., their *covariance*).

Variance

To calculate the covariance, let's first remind ourselves of how we calculate the variance of a single variable. We must take each individual datapoint's difference from the mean, square it, and then sum up all of these squared differences. Finally, we divide the resulting sum of squared differences by N-1. In mathematical notation: $\frac{\sum_{i=1}^N (Y_i - \bar{Y})^2}{N - 1}$

Here is how this works in practice. In the dataset below, the column 'finalgrade' contains individual students' grades on an exam. The next column ('mean') contains the mean grade for the whole group. If we subtract the mean from each grade, we find each student's difference from the mean ('diff'). We then square those differences (column 'sqrdiff').

	ID	finalgrade	mean	diff	sqrdiff
1	1	59	66.15	-7.15	51.18
2	2	71	66.15	4.85	23.49
3	3	62	66.15	-4.15	17.25
4	4	84	66.15	17.85	318.49
5	5	69	66.15	2.85	8.10
6	6	57	66.15	-9.15	83.79
7	7	54	66.15	-12.15	147.72
8	8	64	66.15	-2.15	4.64
9	9	48	66.15	-18.15	329.56
10	10	92	66.15	25.85	668.02
11	11	80	66.15	13.85	191.72

Finally, we sum up all the squared differences in the last column (= 8156.79) and divide by N-1 (8156.79/ 52-1 = 159.92). That is our sample's **variance**. Notice that the variance (159.92) is not on the same scale as the original variable, which was exam grades as percentages – that's because we had to *square* the differences; to

make the variance easier to interpret, we have to bring it back down to the original scale of measurement (Standard Deviation) by finding its root: $\sqrt{\text{variance}} = \sqrt{159.92} = 12.65$, meaning the average person in the sample scored within 12.6 marks of the sample's mean grade. In short:

$$SD = \sqrt{\text{variance}} = \sqrt{\frac{\text{sum of squared differences from the mean}}{\text{sample size} - 1}}$$

Covariance

The covariance is the product of the variances of our two variables. It tells us about the direction of any relationship between two variables:

- positive covariance = the values vary in the same direction
- negative covariance = the values vary in opposite directions

$$\text{covariance} = \frac{\sum_{i=1}^N (Y_i - \hat{Y})(X_i - \hat{X})}{N - 1}$$

In other words, we take each individual participant's score on variable **X** and subtract the mean of X ($\hat{X}$) to find the difference (just like with variance); we do the same for variable **Y**; we then multiply the two differences ($X - \hat{X}$ and $Y - \hat{Y}$) to get their cross product. We sum the cross products of all individual participants and divide by N-1 to get the covariance.

		Y	X	$\hat{Y}$	$\hat{X}$	$Y - \hat{Y}$	$X - \hat{X}$	
▲	ID	finalgrade	submissions	meangrade	meansubmissions	diffgrade	diffsubmissions	crossproduct
1	1	59	3	66.15	7.1	-7.15	-4.1	29.30
2	2	71	6	66.15	7.1	4.85	-1.1	-5.31
3	3	62	9	66.15	7.1	-4.15	1.9	-7.91
4	4	84	10	66.15	7.1	17.85	2.9	51.82
5	5	69	11	66.15	7.1	2.85	3.9	11.11
6	6	57	2	66.15	7.1	-9.15	-5.1	46.65
7	7	54	4	66.15	7.1	-12.15	-3.1	37.63
8	8	64	10	66.15	7.1	-2.15	2.9	-6.25
9	9	48	2	66.15	7.1	-18.15	-5.1	92.51
10	10	92	11	66.15	7.1	25.85	3.9	100.90
11	11	80	11	66.15	7.1	13.85	3.9	54.05

Sum of cross products = 1123

$$\text{Covariance} = \frac{\text{sum of cross products}}{N-1} = \frac{1123}{52-1} = 22.02$$

Person's correlation coefficient (r)

The size of covariance depends on the scale of the variables, which makes it harder to compare the strength of different associations if the underlying variables are in different scales. Pearson's correlation coefficient standardises the covariance as a measure between -1 and 1 , which allows us to assess the strength of the association between variables.

Interpreting correlations

Rules of thumb:

$r=0$ No correlation

$r=.30$ Weak correlation

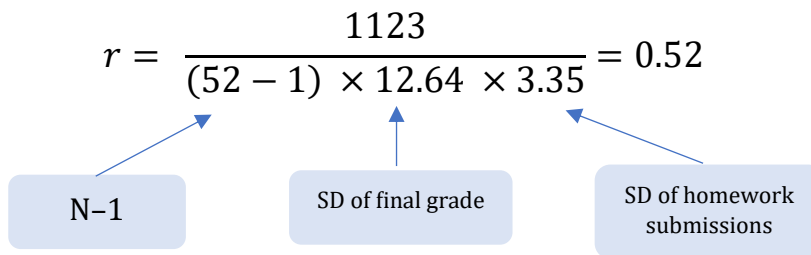
$r=.50$ Moderate correlation

$r=.70$ Strong correlation

$r=1.0$ perfect correlation

$$\text{Pearson's } r = \frac{\text{covariance}}{\text{SDs of the two variables}}$$

In our example:

$$r = \frac{1123}{(52 - 1) \times 12.64 \times 3.35} = 0.52$$


N-1
SD of final grade
SD of homework submissions

Hypothesis testing

Recap: Hypothesis testing

Steps:

1. Determine the null hypothesis
2. Choose a test statistic that if extreme would make us doubt the null hypothesis
3. Simulate or select theoretical null hypothesis sampling distribution
4. Determine where our observed test statistic lies within this distribution and therefore the probability of it being observed if null hypothesis was true.
5. Check the probability (p-value) against a critical threshold (alpha)

r is our sample correlation coefficient. To determine whether we have evidence for a correlation in the population (represented by ρ , the Greek letter "rho") we have to run a hypothesis test

- Null hypothesis (H_0): The correlation between exam grade and homework submission is equal to 0
 - $\rho = 0$
- Alternative hypothesis (H_A): The correlation between exam grade and homework submission is not equal to 0
 - $\rho \neq 0$ (non directional) or
 - $\rho < 0$ (directional) or
 - $\rho > 0$ (directional)

What is the test statistic?

r tells us the strength of the sample correlation (effect size) but what we are interested in is whether this is strong enough to make us doubt that the null hypothesis that the population correlation (ρ) is 0.

This depends on the **effect size** but also the **sample size**. We can therefore calculate a t-statistic that reflects both:

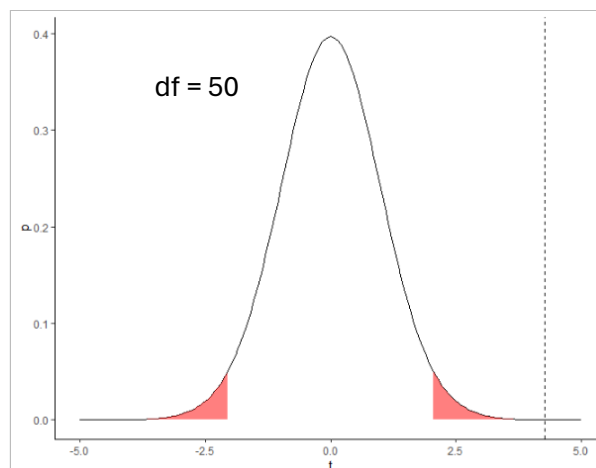
$$t = r \times \sqrt{\frac{n - 2}{1 - r^2}}$$

In our example;

$$t = 0.52 \times \sqrt{\frac{50}{0.74}} = 0.52 \times 8.22 = 4.27$$

Next, we calculate a distribution of potential t-values that we would get if sampling from a population where H_0 is true (a null hypothesis distribution):

The regions shaded in orange show t-values that are likely to appear less than 5% of the time ($p < .05$). Our t-value of 4.27 (represented by a dotted line on the right-hand side of the graph) is even more extreme than that, so it's highly unlikely that we'd find such an extreme t-value if the null hypothesis was true.



Does doing homework make your get a better grade?

The data are consistent with this causal model, but they cannot prove it – **correlation does not equal causation** (<https://www.tylervigen.com/spurious->

[correlations](#)). What other causal models can you think of to explain this association?

Assumptions of Pearson's r

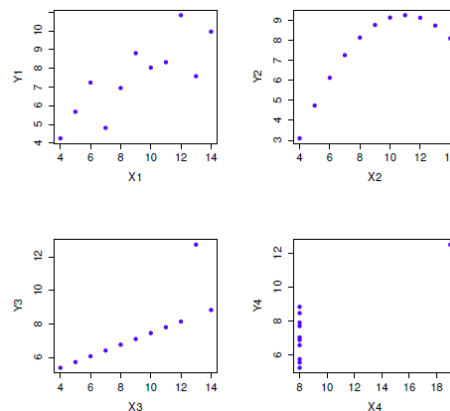
- Interval or ratio level data
- Both variables are normally distributed
- Linear relationship between the two variables
- Homoscedasticity
- No 'bad' outliers – extreme values can distort the relationship between two variables...

Checking assumptions

Create a **histogram** to check for normality.

Create a **scatter plot** to check for:

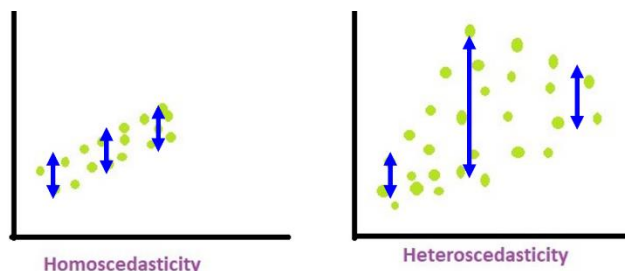
- Linearity
- Outliers
- Homoscedasticity



Anscombe's
Quartet:
 $r = .816$

What is **homoscedasticity**?

Similar amount of variation in y at each value of x . We need the relationship between the two variables to be homoscedastic in order to be able to run the correlation test.



Alternative test: Spearman's rho (ρ or r_s)

If any of the assumptions are not met, you can run **Spearman's** correlation instead. This is the **"non-parametric" version** of a Pearson's correlation coefficient (meaning it doesn't assume normal distribution). We'll learn more about the non-parametric versions of other tests in Term 2.

To calculate Spearman's correlation (ρ or r_s), we simply arrange (rank) the datapoints from smallest to largest and enter the **ranks** of the data points rather than the data points themselves into the correlation equation.

Running a correlation in R

Formula:

```
cor.test(~ variable1 + variable2, data = name_of_dataset)
```

Output:

```
## {r}
cor.test(~ grade + submissions, data = gradehomework)

Pearson's product-moment correlation

data: grade and submissions
t = 4.3022, df = 50, p-value = 7.849e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2876806 0.6942042
sample estimates:
cor
0.5197744
```

Formula for Spearman's rho:

```
cor.test(~ variable1 + variable2, data = name_of_dataset, method = "spearman")
```

Writing up the results:

- Comment on the direction and size of the correlation
- Comment on the statistical significance
- Report correlation coefficient and degrees of freedom in this format $r(\dots) = \dots$
- Report the exact p value, unless it is $< .001$
- Report all values to 2 decimal places

"A Pearson's correlation found homework submission and final grade to be positively correlated with a medium to strong relationship; $r(50) = 0.52$, $p < .001$."